

Raindrop Racers

Target grade: Kindergarten - Grade 2

Time: 35-45 minutes

Team size: 3-4 students

Introduction:

Students investigate how quickly rainwater moves across different types of ground surfaces by building simulated “surface ramps.”

Each team receives several slightly tilted trays or ramps covered with different materials, such as foil, soil, grass, gravel, and sponge. These materials represent surfaces that students may see in their local area including on pavement, bare ground, rocky areas, etc.

Students place water at the top of each ramp and watch as the “raindrops” race toward the bottom then determine which surface creates fast-moving runoff or which surface move water more slowly. The idea is for students to learn how runoff flows across different types of surfaces and which types are impermeable (cannot soak) water.

Learning Objectives

Students will:

- Observe that water moves differently across different surfaces
- Identify which surface absorbs water the fastest and which creates runoff
- Make predictions and compare hypothesis with results

Key Vocabulary

1. Absorb: to soak up water
2. Runoff: water that flows across the ground instead of soaking in
3. Impervious surface: a hard surface that water cannot easily soak through

Materials:

<u>Material</u>	<u>Quantity</u>
Shallow trays, baking sheets, or plastic lids	5
Aluminum foil or plastic sheet	Enough to cover 1 tray
Soil	2-3 cups
Grass, sod, moss, or artificial grass	Enough to cover 1 tray
Gravel or small rocks	2 cups
Sponge pieces or one sponge	Enough to cover 1 tray
Small books or blocks to create a slope	5

Water	~3 cups
Blue food coloring	2-3 drops
Measuring cup or tablespoon	1
Water cups	5
Collection cups	5
Student observation sheet	1 per student or team
Pencils	As needed

Surface Stations:

Prepare five different simulated surfaces:

1. Foil or plastic: Represents pavement, a road, or a parking lot
2. Bare soil: Represents an area without plants
3. Grass or moss: Represents a lawn or a planted area
4. Gravel: Represents a rocky path or gravel area
5. Sponge: Represents a highly absorbent natural area such as a wetland

Teacher Preparation:

1. Cover each tray with one type of material
2. Raise one end of each tray using a book or block (keep each tray at approximately the same angle)
3. Place a collection cup beneath the lower end of each tray
4. Mix a few drops of blue food coloring into the water
5. Measure the same amount of water in the water cups for every test tray

Instructions

1. Show students each of the five test surfaces and ask them to identify where they may have seen each surface in real life (e.g. a road, a sidewalk, a garden etc.)
2. Before testing, have students make predictions on the following:
 - a. Which surface will have the fastest “raindrop”?
 - b. Which surface will soak up the most water?
 - c. Which surface will soak up the least water?
3. Students record their predictions on their handout
4. For each test tray, students place the water cup at the marked starting point at the top and pour the measured water onto the surface
5. Students observe as the water travels down and record whether the water moved **fast, medium, or slow**
6. Observe (and compare to other trays) how much water reached the bottom
7. Have students place the surface types in order from fastest to slowest and least to most water absorbed on their observation sheets
8. Discuss with students at end:

- Which raindrop type “won the race”?
- Which surface held the most water
- Which surface produced the most runoff?
- Which surface might be best near a stream?

Student Observation Sheet

Student Name: _____

Team Number: _____

My Prediction:

I think the fastest raindrop will be on:

- ☐ Foil or pavement
- ☐ Soil
- ☐ Grass or moss
- ☐ Gravel
- ☐ Sponge

I think the surface that will hold the most water is:

- ☐ Foil or pavement
- ☐ Soil
- ☐ Grass or moss
- ☐ Gravel
- ☐ Sponge

Surface	Water Moved Fast, Medium, or Slow	Did Water Soak In?	Did Much Water Reach the Bottom? (Over Half?)
Foil or pavement		Yes/No	Yes/No
Soil		Yes/No	Yes/No
Grass or moss		Yes/No	Yes/No
Gravel		Yes/No	Yes/No
Sponge		Yes/No	Yes/No